

Отчёт к лабораторной работе №1 по предмету «Администрирование в ОС Linux»

execute.sh:

```
#!/bin/bash
set -e
LOGFILE="/home/student/lab1/work3.log"
> "$LOGFILE"
#1
awk -F: '{print "user " $1 " has id " $3}' /etc/passwd >> "$LOGFILE"
#2
chage -l root | grep "Last password change" >> "$LOGFILE"
#3
cut -d: -f1 /etc/group | paste -sd "," - >> "$LOGFILE"
#4
echo "Be careful!" > /etc/skel/readme.txt
#5
useradd -m u1
echo "u1:12345678" | chpasswd
#6
groupadd g1
#7
usermod -aG g1 u1
#8
id u1 >> "$LOGFILE"
#9
if ! id myuser && /dev/null; then
    useradd -m myuser
fi
usermod -aG g1 myuser
#10
getent group g1 | cut -d: -f4 >> "$LOGFILE"
#11
if ! command -v mc >/dev/null 2>&1; then
    apt-get update -y
    apt-get install -y mc
fi
usermod -s /usr/bin/mc u1
#12
useradd -m u2
echo "u2:87654321" | chpasswd
#13
mkdir -p /home/test13
cp "$LOGFILE" /home/test13/work3-1.log
cp "$LOGFILE" /home/test13/work3-2.log
#14
chown u1:u2 /home/test13
chmod 750 /home/test13
chown u1:u2 /home/test13/work3-1.log /home/test13/work3-2.log
chmod 640 /home/test13/work3-1.log /home/test13/work3-2.log
#15
mkdir -p /home/test14
chown 1777 /home/test14
chown u1:u1 /home/test14
#16
cp /bin/nano /home/test14/nano
chown u1:u1 /home/test14/nano
chmod 4755 /home/test14/nano
#17
mkdir -p /home/test15
echo "secret" > /home/test15/secret_file
chown 711 /home/test15
chmod 644 /home/test15/secret_file
#18
echo "u1 ALL=(ALL) /usr/bin/passwd" > /etc/sudoers.d/u1_passwd
chmod 440 /etc/sudoers.d/u1_passwd
```

cleanup.sh:

```
#!/bin/bash
set -e
#1
rm -f /home/student/lab1/work3.log
#4
rm -f /etc/skel/readme.txt
#5
if id u1 >/dev/null 2>&1; then
    userdel -r u1
fi
#6
if getent group g1 >/dev/null 2>&1; then
    groupdel g1
fi
#9
if id myuser >/dev/null 2>&1; then
    userdel -r myuser
fi
#12
if id u2 >/dev/null 2>&1; then
    userdel -r u2
fi
#13
rm -rf /home/test13
#15
rm -rf /home/test14
#17
rm -rf /home/test15
#18
rm -f /etc/sudoers.d/u1_passwd
```

Проверка:

```
[17-02-2026 01:25:16] {a654f287} student@vm-and:~/lab1$ sudo ./execute.sh
[17-02-2026 01:25:37] {a654f287} student@vm-and:~/lab1$ cat work3.log
user root has id 0
user daemon has id 1
user bin has id 2
user sys has id 3
user sync has id 4
user games has id 5
user man has id 6
user lp has id 7
user mail has id 8
user news has id 9
user uucp has id 10
user proxy has id 13
user www-data has id 33
user backup has id 34
user list has id 38
user irc has id 39
user _apt has id 42
user nobody has id 65534
user systemd-network has id 998
user systemd-timesync has id 997
user dhcpcd has id 100
user messagebus has id 101
user systemd-resolve has id 992
user pollinate has id 102
user polkitd has id 991
user syslog has id 103
user uuuidd has id 104
user topdump has id 105
user tss has id 106
user landscape has id 107
user fwupd-refresh has id 989
user usbmux has id 108
user vboxadd has id 999
user student has id 1000
user sshd has id 109
Last password change : Aug 05, 2025
root,daemon,bin,sys,adm,tty,disk,lp,mail,news,uucp,man,proxy,kmen,dialout,fax,voice,cdrom,floppy,tape,sudo,audio,dip,www-data,backup,operator,list,irc,src,shadow,utmp,video,sssl,plugdev,staff,games,users,
nogroup,systemd-journal,systemd-network,systemd-timesync,input,sgx,kvm,render,lxd,messagebus,systemd-resolve,ssh,polkitd,crontab,syslog,uuidd,rdma,tcpdump,tss,landscape,fwupd-refresh,vboxsf,vboxdrmpic,student
uid=1001(u1) gid=1001(u1) groups=1001(u1),1002(g1)
u1,myuser
```

```
[17-02-2026 01:27:46] {a654f287} student@vm-amd:~/lab1$ sudo cat /etc/skel/readme.txt
Be careful!
[17-02-2026 01:30:18] {a654f287} student@vm-amd:~/lab1$ id u1
uid=1001(u1) gid=1001(u1) groups=1001(u1),1002(g1)
[17-02-2026 01:30:34] {a654f287} student@vm-amd:~/lab1$ getent group g1
g1:x:1002:u1,myuser
[17-02-2026 01:30:42] {a654f287} student@vm-amd:~/lab1$ id myuser
uid=1002(myuser) gid=1003(myuser) groups=1003(myuser),1002(g1)
[17-02-2026 01:30:54] {a654f287} student@vm-amd:~/lab1$ id u2
uid=1003(u2) gid=1004(u2) groups=1004(u2)
[17-02-2026 01:31:02] {a654f287} student@vm-amd:~/lab1$ getent passwd u1 | cut -d: -f1,7
command -v mc
u1:/usr/bin/mc
/usr/bin/mc
[17-02-2026 01:31:13] {a654f287} student@vm-amd:~/lab1$ sudo ls -l /home/test13
sudo stat -c "%A %U:%G %n" /home/test13 /home/test13/work3-1.log
total 8
-rw-r----- 1 u1 u2 1309 Feb 17 01:25 work3-1.log
-rw-r----- 1 u1 u2 1309 Feb 17 01:25 work3-2.log
drwxr-x--- u1:u2 /home/test13
-rw-r----- u1:u2 /home/test13/work3-1.log
```

```
[17-02-2026 01:31:31] {a654f287} student@vm-amd:~/lab1$ sudo -u u2 head -n 1 /home/test13/work3-1.log
sudo -u u2 sh -c 'echo test >> /home/test13/work3-1.log' || echo "write denied"
user root has id 0
sh: 1: cannot create /home/test13/work3-1.log: Permission denied
write denied
[17-02-2026 01:31:39] {a654f287} student@vm-amd:~/lab1$ sudo -u u2 head -n 1 /home/test13/work3-1.log
user root has id 0
[17-02-2026 01:33:24] {a654f287} student@vm-amd:~/lab1$ sudo -u u2 sh -c 'echo test >> /home/test13/work3-1.log'
sh: 1: cannot create /home/test13/work3-1.log: Permission denied
[17-02-2026 01:33:34] {a654f287} student@vm-amd:~/lab1$ sudo stat -c "%A %U:%G %n" /home/test14 /home/test14/nano
drwxrwxrwt u1:u1 /home/test14
-rwsr-xr-x u1:u1 /home/test14/nano
[17-02-2026 01:34:02] {a654f287} student@vm-amd:~/lab1$ sudo -u u2 ls /home/test15
ls: cannot open directory '/home/test15': Permission denied
[17-02-2026 01:34:40] {a654f287} student@vm-amd:~/lab1$ sudo -u u2 cat /home/test15/secret_file
secret
[17-02-2026 01:34:48] {a654f287} student@vm-amd:~/lab1$ sudo -u u1 sudo -l
[sudo] password for u1:
Matching Defaults entries for u1 on vm-amd:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin, use_pty
User u1 may run the following commands on vm-amd:
    (ALL) /usr/bin/passwd
[17-02-2026 01:35:10] {a654f287} student@vm-amd:~/lab1$ sudo -u u1 sudo id || echo "command denied"
[sudo] password for u1:
Sorry, user u1 is not allowed to execute '/usr/bin/id' as root on vm-amd.
command denied
```

```
[17-02-2026 01:35:31] {a654f287} student@vm-amd:~/lab1$ sudo ./cleanup.sh
userdel: u1 mail spool (/var/mail/u1) not found
userdel: myuser mail spool (/var/mail/myuser) not found
userdel: u2 mail spool (/var/mail/u2) not found
[17-02-2026 01:37:36] {a654f287} student@vm-amd:~/lab1$ id u1
id: 'u1': no such user
[17-02-2026 01:38:17] {a654f287} student@vm-amd:~/lab1$ id u2
id: 'u2': no such user
[17-02-2026 01:38:20] {a654f287} student@vm-amd:~/lab1$ id user
id: 'user': no such user
[17-02-2026 01:38:23] {a654f287} student@vm-amd:~/lab1$ id myuser
id: 'myuser': no such user
[17-02-2026 01:38:28] {a654f287} student@vm-amd:~/lab1$ getent group g1
[17-02-2026 01:38:38] {a654f287} student@vm-amd:~/lab1$ ls /home
student
[17-02-2026 01:38:49] {a654f287} student@vm-amd:~/lab1$ sudo ls /etc/skel
[17-02-2026 01:38:58] {a654f287} student@vm-amd:~/lab1$ sudo ls /etc/sudoers.d
README
[17-02-2026 01:39:04] {a654f287} student@vm-amd:~/lab1$ ls ~/lab1
cleanup.sh execute.sh
```